KAIZEN MATHS

# Lesson starter

**Volumes of Revolution**

Washer Method

Name

**S1**

Use the washer method with a constant inner radius.

The region between $y = 3x + 5$ and $y = 1$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

**S2**

Use the washer method between two line radii.

The region between $y = 2x + 4$ and $y = 2x + 1$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

**S3**

Use the washer method between a curve and a line.

The region between $y = x^2 + 4$ and $y = 2$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

**S4**

Use the washer method with a constant inner radius.

The region between $y = x + 5$ and $y = 2$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.



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Use the washer method between a curve and a line.

The region between $y = x^2 + 4$ and $y = 2$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

S4

Use the washer method with a constant inner radius.

The region between $y = x + 5$ and $y = 2$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

ALG-3EBB-L3 · S

Show your working

KAIZEN MATHS

# Exit ticket

**Volumes of Revolution**

Washer Method

Name _____

**EA1**

Use the washer method between two line radii.

The region between $y = 2x + 5$ and $y = 2x + 2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

**EA2**

Use the washer method between a curve and a line.

The region between $y = x^2 + 2$ and $y = 2$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.



Show your working

KAIZEN MATHS

Exit ticket

Volumes of Revolution

Washer Method

Name _____

**EB1**

Use the washer method with a constant inner radius.

The region between $y = x + 4$ and $y = 1$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.



Show your working

KAIZEN MATHS

Exit ticket

Volumes of Revolution

Washer Method

Name _____

EA1

Use the washer method between two line radii.

The region between $y = 2x + 5$ and $y = 2x + 2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

EA2

Use the washer method between a curve and a line.

The region between $y = x^2 + 2$ and $y = 2$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

ALG-3EBB-L3 · EA

Show your working

KAIZEN MATHS

Exit ticket

Volumes of Revolution

Washer Method

Name _____

EB1

Use the washer method with a constant inner radius.

The region between $y = x + 4$ and $y = 1$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

ALG-3EBB-L3 · EB

Show your working

# Teacher answers & working

**Volumes of Revolution · Washer Method**

ALG-3EBB-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

## S1

Use the washer method with a constant inner radius.

The region between $y = 3x + 5$ and $y = 1$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = 132\pi$

**Working:**

For a washer, use $V = \pi \int_a^b (R^2 - r^2)\,dx$.

The outer radius is $R = 3x + 5$ and the inner radius is $r = 1$.

Substitute: $V = \pi \int_0^2 [(3x + 5)^2 - 1^2]\,dx$.

Expand and subtract: $(3x + 5)^2 - 1^2 = 9x^2 + 30x + 24$.

Integrate: $\int (9x^2 + 30x + 24)\,dx = 3x^3 + 15x^2 + 24x$.

Evaluate: $\left[3x^3 + 15x^2 + 24x\right]_0^2 = 132$.

Therefore $V = 132\pi$.

## S2

Use the washer method between two line radii.

The region between $y = 2x + 4$ and $y = 2x + 1$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = 21\pi$

**Working:**

Use the washer formula $V = \pi \int_a^b (R^2 - r^2)\,dx$.

On this interval, the outer radius is $R = 2x + 4$ and the inner radius is $r = 2x + 1$.

Set up: $V = \pi \int_0^1 [(2x + 4)^2 - (2x + 1)^2]\,dx$.

Expand both squares and subtract: $(2x + 4)^2 - (2x + 1)^2 = 12x + 15$.

Integrate: $\int (12x + 15)\,dx = 6x^2 + 15x$.

Substitute the bounds: $\left[6x^2 + 15x\right]_0^1 = 21$.

Therefore $V = 21\pi$.

# Teacher answers & working

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## S3

Use the washer method between a curve and a line.

The region between $y = x^2 + 4$ and $y = 2$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = \dfrac{223\pi}{15}$

**Working:**

Use $V = \pi \int_a^b (R^2 - r^2)\,dx$.

The outer radius is $R = x^2 + 4$ and the inner radius is $r = 2$.

Set up: $V = \pi \int_0^1 [(x^2 + 4)^2 - 2^2]\,dx$.

Expand and simplify: $(x^2 + 4)^2 - 2^2 = x^4 + 8x^2 + 12$.

Integrate: $\int (x^4 + 8x^2 + 12)\,dx = \dfrac{1}{5}x^5 + \dfrac{8}{3}x^3 + 12x$.

Evaluate: $\left[\dfrac{1}{5}x^5 + \dfrac{8}{3}x^3 + 12x\right]_0^1 = \dfrac{223}{15}$.

Therefore $V = \dfrac{223\pi}{15}$.

## S4

Use the washer method with a constant inner radius.

The region between $y = x + 5$ and $y = 2$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = 117\pi$

**Working:**

For a washer, use $V = \pi \int_a^b (R^2 - r^2)\,dx$.

The outer radius is $R = x + 5$ and the inner radius is $r = 2$.

Substitute: $V = \pi \int_0^3 [(x + 5)^2 - 2^2]\,dx$.

Expand and subtract: $(x + 5)^2 - 2^2 = x^2 + 10x + 21$.

Integrate: $\int (x^2 + 10x + 21)\,dx = \dfrac{1}{3}x^3 + 5x^2 + 21x$.

Evaluate: $\left[\dfrac{1}{3}x^3 + 5x^2 + 21x\right]_0^3 = 117$.

Therefore $V = 117\pi$.

# Teacher answers & working

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ALG-3EBB-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

## EA1

Use the washer method between two line radii.

The region between $y = 2x + 5$ and $y = 2x + 2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = 66\pi$

**Working:**

Use the washer formula $V = \pi \int_a^b (R^2 - r^2)\,dx$.

On this interval, the outer radius is $R = 2x + 5$ and the inner radius is $r = 2x + 2$.

Set up: $V = \pi \int_0^2 [(2x + 5)^2 - (2x + 2)^2]\,dx$.

Expand both squares and subtract: $(2x + 5)^2 - (2x + 2)^2 = 12x + 21$.

Integrate: $\int (12x + 21)\,dx = 6x^2 + 21x$.

Substitute the bounds: $\left[6x^2 + 21x\right]_0^2 = 66$.

Therefore $V = 66\pi$.

## EA2

Use the washer method between a curve and a line.

The region between $y = x^2 + 2$ and $y = 2$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = \dfrac{23\pi}{15}$

**Working:**

Use $V = \pi \int_a^b (R^2 - r^2)\,dx$.

The outer radius is $R = x^2 + 2$ and the inner radius is $r = 2$.

Set up: $V = \pi \int_0^1 [(x^2 + 2)^2 - 2^2]\,dx$.

Expand and simplify: $(x^2 + 2)^2 - 2^2 = x^4 + 4x^2$.

Integrate: $\int (x^4 + 4x^2)\,dx = \dfrac{1}{5}x^5 + \dfrac{4}{3}x^3$.

Evaluate: $\left[\dfrac{1}{5}x^5 + \dfrac{4}{3}x^3\right]_0^1 = \dfrac{23}{15}$.

Therefore $V = \dfrac{23\pi}{15}$.

# Teacher answers & working

**Volumes of Revolution · Washer Method**

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## EB1

Use the washer method with a constant inner radius.

The region between $y = x + 4$ and $y = 1$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = 90\pi$

**Working:**

For a washer, use $V = \pi \int_a^b (R^2 - r^2)\,dx$.

The outer radius is $R = x + 4$ and the inner radius is $r = 1$.

Substitute: $V = \pi \int_0^3 [(x+4)^2 - 1^2]\,dx$.

Expand and subtract: $(x+4)^2 - 1^2 = x^2 + 8x + 15$.

Integrate: $\int (x^2 + 8x + 15)\,dx = \frac{1}{3}x^3 + 4x^2 + 15x$.

Evaluate: $\left[\frac{1}{3}x^3 + 4x^2 + 15x\right]_0^3 = 90$.

Therefore $V = 90\pi$.